



Easy Digital Twin Toolkit

User Manual - Version 0.9



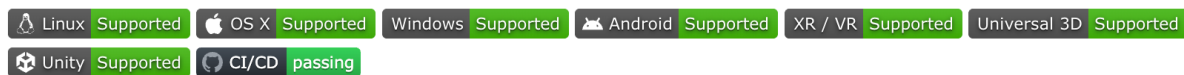
**Gemeente
Rotterdam**

1. About the Toolkit

Welcome to the manual for the Digital Twins Toolkit. This toolkit makes it easy to bring a Digital Twin (of Rotterdam) into your Unity projects and interact with it.

The Toolkit has the following features which make it easier to start developing an XR application with live City data:

- A User Manual with setup, usage and deployment guide.
- Integration of Data Catalogues for browsing selecting and importing Datasets and Digital Twins.
- Automatic Installation of Dependencies and Cesium for Unity for rendering 3D meshes from the City Data.
- Automatic Scene, Georeference and Cesium for Unity configuration.
- Plug-and-play custom made light weight controllers prefabs with gamepad support.
- Example scenarios and scripts to get you started.
- Optimisation and configuration tools for performance and deployment.



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Publication and Releases

Clearly Hub: <https://hub.clearly.app/apps/6a2b147ff03efa527b952d73/details>

Unity Package (UPM): <https://github.com/Basisinformatie/easy-digital-twins-in-unity.git>

Documentation

Setup Guide (SETUP_GUIDE.md): How to install and configure the toolkit. User Interface Guide

(USER_GUIDE.md): A concise overview of the buttons, windows, and workflows.

Licensing

This toolkit includes custom content, assets, and source code released under the MIT License and Creative Commons licensing terms. For full details, please refer to the official LICENSE file included in the repository.

In short:

- Please credit the toolkit when you use it, use the proper attribution methods.
- Commercial use is Permitted.
- The authors and owner are not responsible or liable for any outcomes resulting from its use.
- Do not use the toolkit for unethical purposes.

Credits and Attributions

- Helicopter 3D Model: Created by pravnav27 (CC Attribution). Source: <https://sketchfab.com/3d-models/helicopter-c33f1be4708b422e8750d9b1db9894bf>
- Car for Games Unity 3D Model: Created by Unity (shehab house) (CC Attribution). Source: <https://sketchfab.com/3d-models/car-for-games-unity-4d9bbde680fe41349d7ad2b4672a720b>
- Cesium for Unity: <https://cesium.com/learn/ion/content-usage-and-attribution-guide/>

2. Setup Guide

a. Prerequisite

Unity Engine The toolkit is optimized for Unity 6 (version 6000.3.9f). While it is compatible with earlier and probably later versions, some implementation steps (for example, deployment to specific platforms) may vary. Download Unity Hub: <https://unity.com/> Unity Hub installation support documentation: <https://docs.unity.com/en-us/hub>

Git (Recommended) While not strictly required, using Git is highly recommended for easy updates and version management within your project. Download Git: <https://git-scm.com/install/>

b. Installation

The toolkit is designed to work with the following Unity templates:

- Universal 3D (URP/SRP)
- AR Mobile
- VR
- Multiplayer variants of the above

Adding the Package:

1. Open your Unity project.
2. Open the Package Manager (Window > Package Manager).
3. Click the plus (+) button in the top-left corner.
4. Select "Add package from git URL...".
5. Enter the following URL: <https://github.com/Basisinformatie/easy-digital-twins-in-unity.git>
6. Wait for the installation to complete.

Alternatively, you can download the zip from the npm branch from the repository on github, unpack it, and import the package manually.

c. Initial Launch

Once installed, a new menu item will appear in the Unity top menu:

1. Navigate to Rotterdam Digital Twins > Launch UI.
2. The Toolkit Window will open and automatically begin installing necessary dependencies, such as Cesium for Unity.
3. Note on popups: You may see warnings regarding a "scoped registry" or a "missing signature" related to the Cesium plugin. These are expected and can safely be disregarded and closed.
4. Once the setup is complete, the loading animation will stop, the Main Menu functions will be unlocked and you can begin using the toolkit.

d. Hardware and Storage Requirements

Depending on your target platform (Web, AR, VR, etc.), additional modules may be required.

Development and Build Space

- Standard Development: less than 5 GB
- VR Multiplayer Module: approximately 25 GB
- AR Multiplayer Module: approximately 12 GB
- iOS Deployment (Xcode and temporary storage): approximately 20 GB

Runtime Storage (3D Tiles and Caching) - The toolkit uses a 3D Tile system (via Cesium) to stream high-detail environments.

- Digital Twin Loading: Averages between 2 GB and 8 GB of temporary disk space, depending on the resource complexity.
- Mobile / Standalone Optimization: For devices with limited memory (for example, the Meta Quest 3 with a 512 MB cache), the toolkit includes performance optimization settings. For complex applications, you may need to manually tune the mipmap configuration, caching, and tile loading to fit within your device's limits.

3. Usage Guide

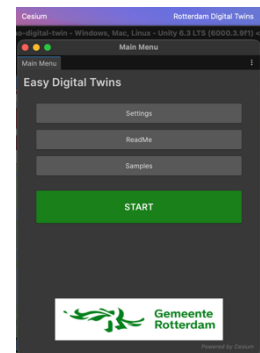
a. Toolkit UI Overview

This section provides an overview of the "Easy Digital Twins" toolkit interface in Unity.

Main Menu

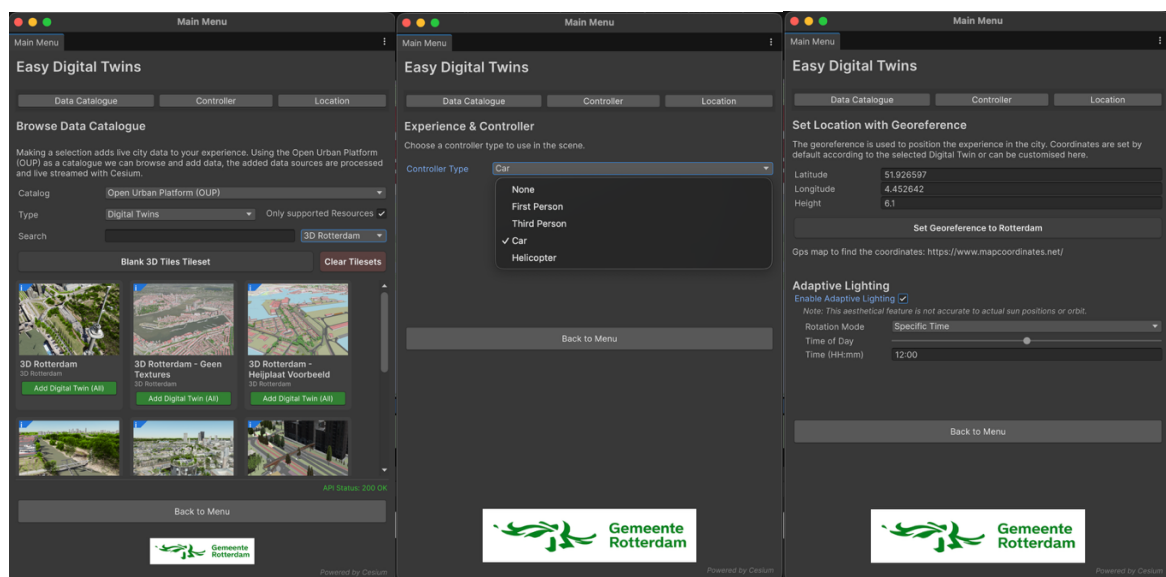
The entry point for the toolkit, accessible via Rotterdam Digital Twins > Launch UI.

- START: Opens the Shopping Wizard to configure your scene.
- Settings: Configure project detection, graphics presets, and web deployment.
- ReadMe: View the project's documentation directly in Unity.
- Samples: Explore sample scenes and implementations.



Shopping Wizard (Tabs)

After pressing START, the Shopping Wizard contains three tabs for switching between the main functionalities of the Toolkit.



Data Catalogue

Search and add live city data from the Open Urban Platform (OUP).

- Catalog Dropdown: Switch between different data providers (for example, OUP or Mock).
- Type Dropdown: Filter between "Digital Twins" and "Datasets". A Digital Twin is a collection of resources with metadata and preconfigured by the owner. A Dataset contains a specific resource, sometimes options for different variants of resources within the same dataset are offered.
- Search Field: Find data by title, description, or tags.
- Hub / Category Dropdown: Filter results by specific organizational hubs or categories.
- Only Supported Resources: When enabled, hides entries that do not have compatible 3D formats (3D Tiles, WMS, etc.).
- Add to Scene: Click on a result to instantly add it to your Unity scene.

Controller

Select how you want to navigate your Digital Twin.

Controller Type — choose between:

- None: Uses a standard Main Camera.
- First Person: Standard walking character.
- Third Person: Character with a following camera.
- Car: Driveable vehicle controller.
- Helicopter: Flyable helicopter controller.

Note: Selecting a new type automatically replaces the existing controller in your scene and adds a platform that prevents the controller from falling through the environment during loading.

Location

Define where your experience is situated in the world.

- Latitude / Longitude / Height: Manually set the georeference coordinates.
- Set Georeference to Rotterdam: Quickly reset the location to the center of Rotterdam.
- Adaptive Lighting: Enable dynamic lighting based on time.
 - o Rotation Mode: Choose between "Specific Time" or "Full Day Cycle".
 - o Time Slider: Manually adjust the time of day.

Settings

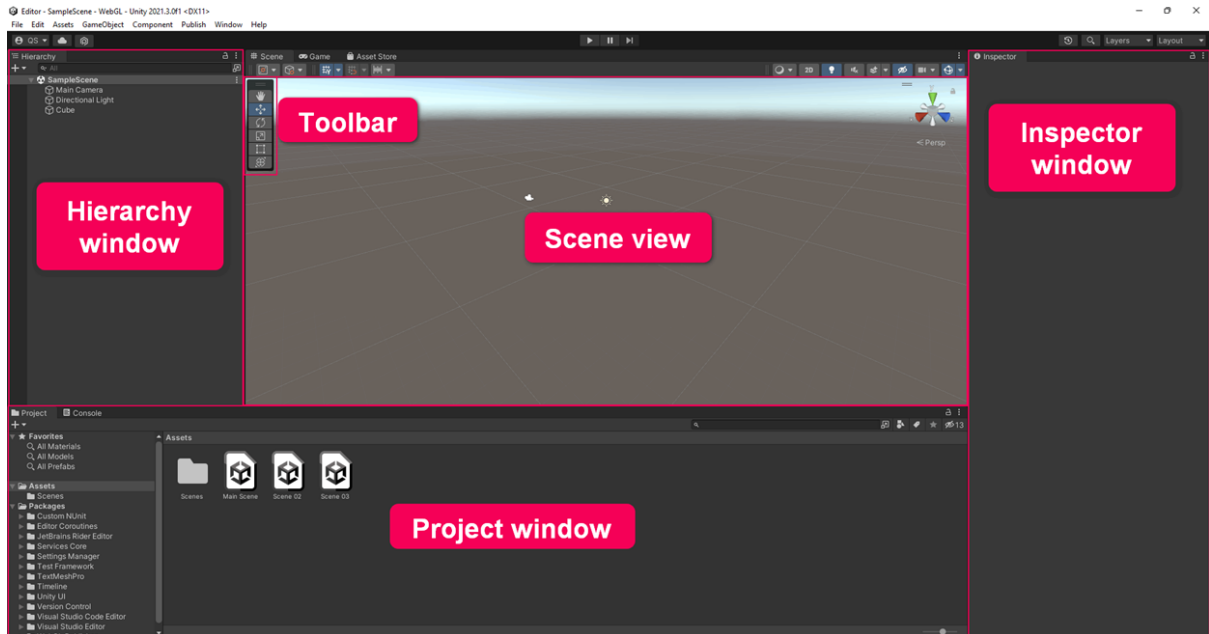
- Project Detection: Automatically detects whether you are targeting VR, Web, or Mobile.
- Graphics and Performance: Apply presets (Low, Medium, High) to optimize 3D Tileset loading and camera clipping.
- Web Deployment: Specific optimizations for WebGL builds.

b. Step by Step example usecase

Let's get to know the toolkit by making a simple 3D flyer experience.

Step 1. Press START in the toolkit window and let's take a look at the Data Catalogue:

When we open the Shopping Wizard, through pressing START, it opens the first tab which is the Data Catalogue. By default the selected Catalog is set to Open Urban Platform, Type is set to Digital Twins and the checkbox is set to show Only supported Resources. After a short loading duration we can browse the available results, each result is displayed in a Card Component containing: a preview if available, blue information icon which directs you to the clearlyhub page, the name, the publishing hub and a button to add.



Step 2. Lets add our first Digital Twin to our experience:

Filter on hub and set it to Hub van Joris Koolen.

Locate the result Bombarie op de Boompjes and press the green Add button.

In the Hierarchy Window a CesiumGeoreference object has been created. This object contains more objects, these are Cesium 3D Tilesets which build the different resources or layers that make up the meshes of the Digital Twin.

Note: When developing it is advised to only configure the Georeference and 3D tiles, when making or adding your own objects do this outside/seperate of the CesiumGeoreference object hierarchy.

Step 3. Removing and adding individual resources:

In the Scene View we can see that the Terrain is not loaded correctly, this can have a multitude of reasons – for now we will delete it.* Selecting the Terrain 3D Tileset we can uncheck it in the Inspector Window or simply delete it. We can repeat this step for the Tileset with the Trees.

*(*note: this Digital Twin also contains a topographical terrain element, BGT which will act as our Terrain instead)*

Reopening the Data Catalogue, set Type to Datasets, set the hub to 3D Rotterdam and locate the dataset 3D Rotterdam – Bridges, this has two resources Add one of them.

Step 4. Adding a controller:

Now that the experience is populated with a map, buildings, bridges and some custom shapes we will want to move through it using a controller. Navigate to the Controller tab in the Shopping Wizard. Select the Helicopter in the Controller Type dropdown menu. This removes the Camera object, places a StartingPlatform and places a Helicopter-Rig prefab which contains a new Camera object.

- By default a Camera with a MainCamera tag is always necessary to load 3D tiles correctly. Based on radius around it and looking distance the 3D tilesets are visualised dynamically.
- A StartingPlatform is placed so the controller doesn't fall through the floor during 3D tile loading. It contains a Ground Snap script which places it on top of the mesh/floor below it.
- The Helicopter-Rig is placed just above the StartingPlatform. Make sure to double check no clipping occurs and manually reposition if necessary. This controller has keybindings which can be found and adjusted in the Custom Helicopter Controller script or directly in the Inspector Window.

Optional: You can add an extra feature to the Helicopter-Rig that show building addresses or bridge names. In the Inspector Window press add component or simply drag and drop the script Building Selector on the object.

When a controller/rig is added through the toolkit it will remove any objects it

Step 5. Georeferencing and Location

Navigate to the Location tab in the Shopping Wizard. Here you can change the gps coordinates where your experience should take place through Latitude, Longitude and Height. When you have added a Digital Twin the location is usually already set but you can overwrite this to your liking. There is also a Set Georeference to Rotterdam Button which takes you to a default location op Laan op Zuid. Using a gps website like www.mapcoordinates.net you can pick a location on a map and configure them yourself.

With the adaptive lighting feature checkbox checked it adds a simple skybox with directive lighting to the environment. Now you can select a specific time or make it continuous.

Note: This is an optional aesthetical feature and not accurate to the actual sun position timings or orbit.

Step 6. Testing our experience in the editor

Pressing the play button above the Scene view lets you run the experience and fly through the city.

Controls are with WASD, SPACE for more power and SHIFT for less. Alternatively you could use a gamepad.

When you have added the Building Selector you can also click on a building or a bridge to show more information about it.



A demo of a similar experience made with the Toolkit deployed for web can be tried here: <https://hub.clearly.app/apps/6a2ffd3d6b336761f2c4aba4/details>

c. Samples and Extra Features

The toolkit comes with a variety of small features and samples that are not part of the core functionality. However, they can be fun to use or useful as a starting point to build upon. Some of these features are meant to showcase how something is implemented, giving you a better understanding of what is possible and how to write your own scripting. In some cases they may be directly useful for your own usecase. All assets can be found in Easy Digital Twin Toolkit Folder in the Project Window.

Included sample features:

- A functional Boat controller, made through a small adjustment on the Car controller.
- Showing building data in a UI.
- Painting on buildings.
- Deleting meshes.
- Ground snap.
- Teleportation on 3D Tiles (for VR).
- Y-axis mover script.
- Controller support and remappable keys.

Boat controller

A simple boat controller is provided, based on a model adjustment of the Car controller. You can find it in the folder: Runtime > Prefabs > Features > Boat. The prefab is called CustomBoatControllerRig and can be used as-is. It is a standalone prefab that can be dragged straight into the Hierarchy or Scene window.

Building Selector

The BuildingSelector is a script that shows address information or the name of bridges in a UI when the user clicks with the mouse. It can be added as a component to a controller.

Mesh Painter

The MeshPainter lets users draw or paint directly onto buildings. It can be added as a component to a controller.



Building Deleter

The Building Deleter makes it possible to make meshes disappear at runtime. It can be added as a component to a controller.

Ground Snap

When added as a component to an object, the Ground Snap script will place that object on top of the surface beneath it (with a small offset).

Cesium Teleporter

The landscape can be made teleportable for VR navigation by adding the custom made Cesium Teleporter component.

Y-Axis Mover

A script that lets you dynamically raise or lower an object along the Y-axis — for example, to change the water level at runtime.

Gamepad and joystick support

Gamepad and joystick support is integrated into the custom controller, alongside the standard XR and keyboard-and-mouse controls. You can remap keys directly in the UI, or check the scripts for more options. For controller support in web builds, I recommend following a dedicated guide, since it works slightly differently.



Cesiumlon

Although not a feature of the Toolkit itself, Cesiumlon is compatible. Meaning you can import maps and different tilesets from google, bing and other open services using Cesiumlon.

4. Deployment and Running the Experience

Besides running an application and testing the runtime in the Unity Editor, the toolkit is also compatible with building and deploying to different devices.

Integration for the following (types of) systems:

- 3D application for the operating systems Windows, Mac, Linux and Android
- AR App with Google Play Services for ArCore with Android and iPadOS
- Virtual Reality experience with OpenXR for, among others, Android VR
- Web App using WebAssembly with WebGL.

Deployment compatibility has been tested successfully for the following types of devices:

- Handheld
- Smartphone and Tablet
- Desktop
- VR headset
- Website

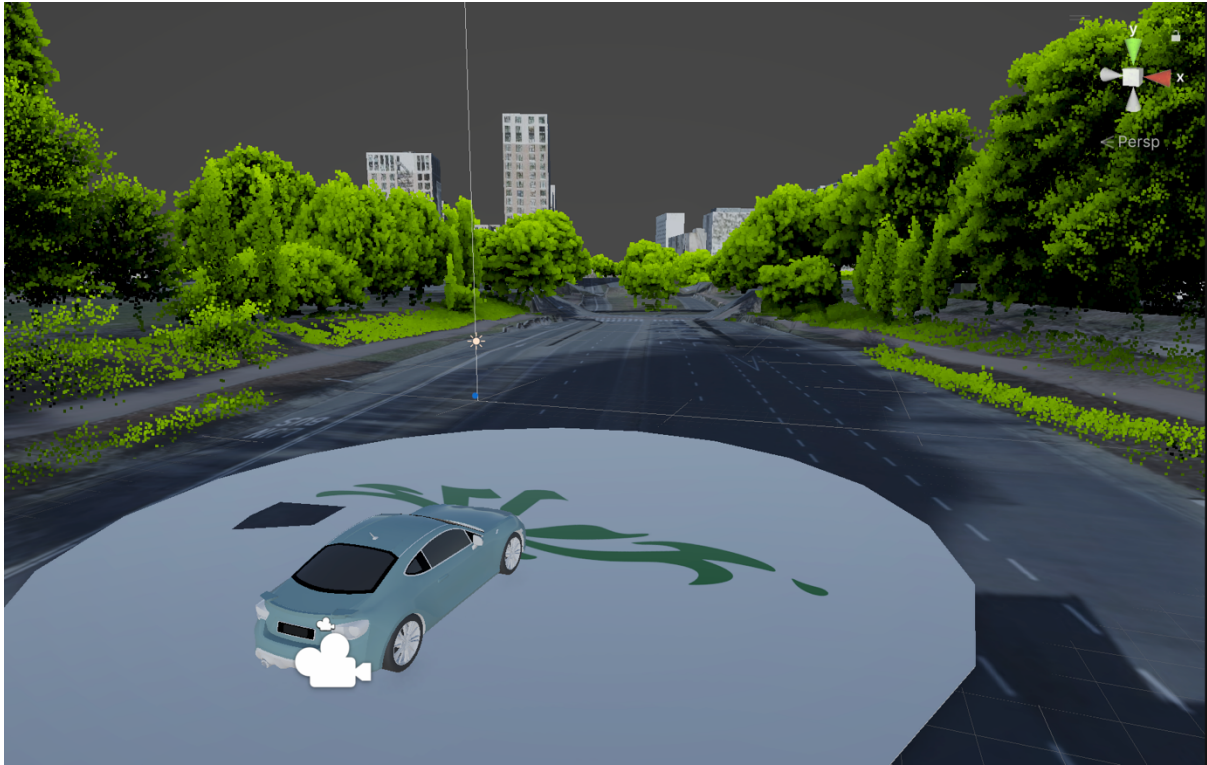
Compatibility and setup

Built into the toolkit is a helpful indication as to what type of experience your project scene is currently compatible with. It decides this based on installed packages and platform deployment kits. Simply go to Settings menu from the Main menu in the toolkit and view the Project Detection segment:

- The project Type tells you if the project is primarily suited for 3D universal, AR or VR applications.
- Compatibility indicates for what operating system your project is suited.

For development, experience design and controls it is still of course necessary to implement controls that suit the type of experience or use the native controls set of the type of experience or device. The toolkit comes with controllers that work with mouse keyboard and gamepad, but by design it does not contain touchscreen or VR sensor and controller support. Instead, we rely on using the embedded prefabs that use the native controls for a type of device that come with Unity, namely the ones that are provided through the template projects. You can also implement your own.

It is advised, using Unity Hub, to get and set the Engine version to 6000.3.9f1 and add the modules necessary for the device you wish to deploy to. Based on the type of experience you want to build you want to have the right packages installed, manually this can quickly become complicated, that's why we suggest using the template available from Unity Hub when starting a new project.



3D Universal - Desktop app and web

For deployment to desktop app and web. Make a new project with the Universal 3D (SRP) template or use an existing SRP compatible project.

This is pretty much setup as plug-and-play:

- Use the Toolkit as normal adding datasets, controller and configuring the location.
- If you intend for web deployment use the Settings menu from the Main menu in the toolkit and click the Configure for Web Deployment button. *Note: this configures the project for WebGL Wasm, enabling multi thread support and increase the browser memory size allocation to 1024mb.*

Move on to the Build and Build and Run chapter.

VR - Virtual Reality

For deployment to VR headsets. Start a new project with the VR template or use an existing VR compatible project.

The VR template comes with a preconfigured scene which contains a lot of things. I would suggest tidying up the Hierarchy/Scene from everything that is not interesting or needed. Most important is to keep the XR Origins hands which is our VR Controller Rig and maybe you can leave for example the interactable objects and tutorial elements.

You can now use the toolkit as usual and build your experience, with one exception, we already have a working controller now so the Shopping Wizards Controller tab should not be used.

The VR controller has a few types of locomotion available e.g. climbing, flying, teleporting and walking. The meshes generated by the 3D tiling system are not ready for teleporting by default,

for this you will need to add the Cesium Teleporter script to the Cesium 3D Tileset object you wish the VR teleportation to work on.

By default Fog is kind of intense which is not preferable on city scale environments, use the Unity top menu > Windows > Rendering > Lighting. Navigate to the Environment tab and lower or remove the Fog settings.

After these steps you can Build your VR experience and transfer the apk to the VR-headset for local installation.

If you have the VR headset connected to your device with the Unity Editor and if it is detected as such - it is easiest to use the Build and Run option.

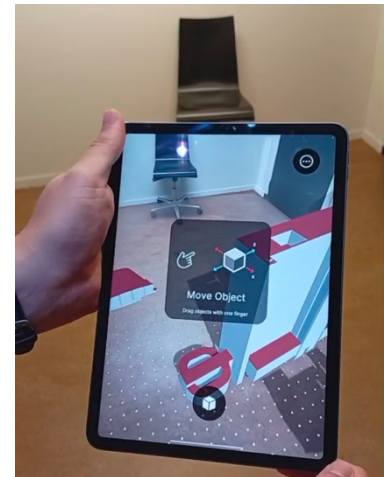
AR - Augmented Reality

For deployment to smartphone and tablet with ARCore. Start a new project with the AR Mobile template or use an existing AR compatible project.

Use the Shopping Wizard Data Catalogue to add some datasets or Digital Twin. Save the created Georeference as a prefab in the designated Asset folder, Assets> MobileARTemplateAssets>Prefabs, in the Project Window and remove the Georeference from the Hierarchy Window/Scene View.

Use the Settings menu from the Main menu in the toolkit and click the Integrate Cesium into AR button. *Note: This will add a custom option in the object interface of the application that allows you to place the Georeference.prefab down during the runtime.*

For Android you will need to turn on development mode on the device.



For iPhoneOS and iPadOS deployment you will need to create a Xcode developer account, setup an Xcode environment with the right SDK for your device and enable Developer mode on your Ipad. Make sure to have enough storage space on both of your devices.

ARCore compatibility and devices: <https://developers.google.com/ar/devices>

Move on to the Build and Build and Run chapter.

Multiplayer for Universal and VR

Unity Hub offers a variety of multiplayer templates which are compatible with the toolkit: 1. “Multiplayer Third Person Gameplay”, 2. “VR Multiplayer” and 3. “Multiplayer FPS”.

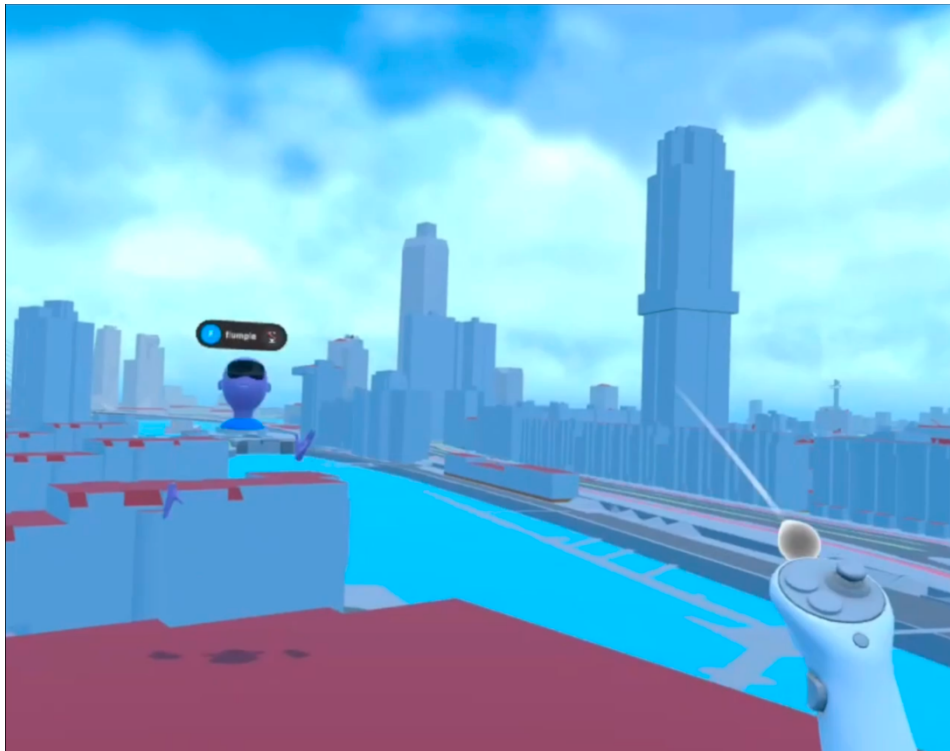
For using any multiplayer template we in principal advise the same points as the “VR – Virtuality” chapter.

- The templates come with their own functional controllers, use that one. These controllers are suited for the type of experience and are network/multiplayer ready.
- The template is full of components, often an arena like playing field/level. Decide which components to keep and which to remove. Often the spawn points are necessary for new players to join the experience so keep and position those well.

- Check any lighting, fog and environmental options and adjust/customize them as desired.

For universal templates, template 1 and 3, when you play the app you get the options to host a game or join an existing one. There are some variants and based on the version of the templates the process might differ a little – one player can host or create a game and the other players can join either by a quick join or by entering an ip and port. The network manager in the editor can let you decide on a port and you will have to check the ip-address of the hosting device. (e.g. check your device network configuration, or, use “ipconfig” in command prompt for windows and check the ipv4 address, or, use a whatsmyip website - could be less reliable in a complex network.) Mostly this is not complicated and plug and play.

Note: There are also options to host and deploy the app to a server with settings and selecting the right platform – this will be a host of the session that will always be up with a world but without a playercontroller. Check the official documentation of the template of your usecase for more information on this.



Setting up a VR multiplayer requires a few steps to get it to work, here we will describe some steps which are required to set it up:

Login in Unity website and make sure you have created an Organization.

<https://cloud.unity.com/acquire>.

- In the organizations Unity Cloud Dashboard and setup the Unity Cloud Gaming Service (UGS). Make sure the Unity Project is linked to the Unity Cloud. In the Unity Editor go to Edit > Project Settings and go to Services:
 - o Select your Organization
 - o Select use an existing cloud project or select create a new one if the VR multiplayer project is not yet available in the cloud.
 - o Click the button Create new cloud project and link to Unity project.
 - o Connect to Vivox Voice Chat Service – go to Project settings > Services > Vivox.

- On the website on the Unity Cloud Dashboard go to Projects and select your VR Multiplayer Project. On the Services Tab make sure “Lobby”, “Relay” (for peer to peer) and “Vivox Voice and Text Chat” are added.
- Optional: using an Authentication Provider service you could select an identity provider for adding authentication to your experience. (Google, Steam, username password, etc.)

“Build” and “Build and Run”

In Unity’s top menu navigate to File > Build Profiles.

Select the preferred Platform and press the Switch Platform button to make it the active platform. Wait patiently for the required kit to have loaded in, note: there is also the in-editor status bar located in the bottom right.

Once the platform is set to active and Unity is ready: you can now select Build or Build And Run depending on your usecase.

Select a folder where to build to, give it a name and watch out for errors!

Note: Use the Toolkit Settings Menu to quickly optimize the performance suited for your usecase if you are having trouble configuring performance.

5. Troubleshooting

If you are in unfamiliar territory and try to create complex things too fast you will be faced with complex stubborn roadblocks. Through experience I advise you to read the documentation carefully and double check your steps. Make sure you have followed the steps thoroughly, you are using the right packages and setup and read the error messages as to get an indication of what is going wrong! Sometimes the best course of action to take is to retrace/undo your steps until you get back to a point where it did work - or try to test after every change you make to prevent problems in the first place.

The Toolkit is here to help: in it is implemented a lot of debug messages, userfeedback and error features that should tell you what is going on. We also have a whole section down below with frequently occurring problems and solutions. Sadly it is also very possible you are encountering problem that is neither your or the toolkits fault. Good luck!

Frequently occurring errors and fixes:

- **Active input handling and input mode Error**
 - o Go to the Unity top menu and navigate to Edit > Project Settings. Go to Player, select the tab with platform icon matching your usecase and find the Active Input Handling setting and set it to Both.
- **Api not reachable, toolkit Data Catalogue window shows a red Api status**
 - o Check for Errors or debuglogs for indication of what is going on. Check for a working internet connection. Navigate to the clearly hub website. Navigate to the dataset matching your usecase. Try to reach the source link with the browser. If everything works, the API service is likely under maintenance.
- **Dataset not showing (correctly):**
 - o Use the Only supported Resources settings in the filter menu on Data Catalogue. Make sure from the source that the dataset is in the supported format and the resource is in working order. A lot of the offered datasets are

tests or have been taken offline (some are not being managed). There is a known issue with i3dm datasets, this will hopefully be resolved in the future. Make sure you have enough storage on your device. Pointcloud have an Annotation option. If it is about a WMS, this needs to be projected onto a suited Cesium 3D Tileset, try to re-add it. Check for a stable, fast and working internet connection.

- **Toolkit loading screen seems stuck**
 - Check for a working internet connection. Check for Errors. It is possible that Cesium version is not compatible with the unity version.
- **Can't find the VR Headset**
 - Detach the VR headset. Turn debugging mode off and on on the VR headset. Reconnect the headset using a VR suited high speed data cable. If it is about a Mac connection to a VR headset, this requires the meta headset to be in developer mode. Make sure to trust and accept any file transfer or debugging prompt either/both devices give. Make sure the platform you build and run to match. If all else fails try to build and file transfer another way and install the apk on the device locally.
- **Crash Of the application after successful deployment**
 - Check the capacity of the chosen hardware/device and the performance the application demands. Consider caching for data you want to stream, application size, and graphical limitations (shaders, anti-aliasing, resolution, etc.). Use the performance presets in the Settings menu from the Main menu in the toolkit and work from indicated there. There may be a specific dataset that is corrupt or too large. Try adding datasets one by one and testing them. Textured tiles, WMS and pointclouds can ask a lot of memory and performance, consider lowering the quality or using different tilesets.
- **Missing Packages**
 - Use the toolkits Settings menu for an indication of what is going wrong. Check if you are using the right version of unity and the right templates. Did you add any libraries or additional content from the store, it might be a dependency issue.
- **Slow loading times**
 - Check for a stable internet connection. WMS can load slow by nature. Check if the device has the capacity to meet the app demands. Use the Toolkit's Settings menu and use the Performance preset. Textured tiles, WMS and pointclouds can ask a lot of memory and performance, consider lowering the quality or using different tilesets.
- **Null pointer reference or script errors.**
 - Try to read the error, or retrace your steps by removing and re-adding elements one at a time to see where the problem originates.
- **Cesium Ion or API token error**
 - This happens because of an invalid token with Cesium. It usually refreshes or creates a new token after the error. Try clearing the error - if it persists, take a look at the Cesium documentation. A Cesium Ion API token or account is not required unless you want to use Cesium Ion (using Google Maps, Bing, etc.) in Cesium for Unity.
- **Cesium general errors**
 - Check whether the Cesium menu item at the top is available if not the Cesium install was unsuccessful. Read the errors to get an indication - if there are many it is likely that the current version of your Unity Editor is not compatible with your installed Cesium for Unity version.
- **Graphical glitches**

- This is a known issue with Point clouds, try switch animation on or off in the Inspector Window. The vegetation point cloud is up to 8.5 GB of data, which can also lead to problems with continuous loading and recharging. A round shadow can appear under the controller, this has to do with the rendering technique and graphics card – can sometimes be solved by not using pointclouds. If things are and remain purple or blue you are likely dealing with incompatible materials – Materials behave differently in SRP/URP and HDRP.
- **Helicopter-Rig is hard to control (frictionless)**
 - When the helicopter controller is hard to control and glides away at high speeds it is likely you are running a unity editor with a different version than advised. Adjust the Rigidbody component on the Helicopter-Rig – adjust the Linear Damping to 1 and Angular Damping to 4.

Also make sure to consult the respective manuals based on which aspect of the system it is related to:

- Cesium for Unity: <https://cesium.com/learn/unity/>
- Clearly hub/Open Urban Platform: <https://hub.clearly.app/connect/technical-overview>
- Unity: <https://docs.unity3d.com/6000.3/Documentation/Manual/index.html>

Have you tried turning it off and on again?



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